

5CSE 370 – Database Systems

Assignment 02 | Sample Solution

Just provided a short answer. You obviously need to draw the schema diagram with proper FDs..

Question 1 [CO5] : 10 Points

Consider the following relation:

CAR_SALE (Car, Salesperson, Commission, Date_sold, Discount_amt)

The primary key of the relation is underlined. Assume that a car may be sold by multiple salespersons and so {Car, Salesperson} is the primary key.

Suppose the following additional dependencies exist:

FD1 : Date_sold $\rightarrow$ Discount_amt

FD2 : Salesperson $\rightarrow$ Commission

Based on the given primary key,

(i) Explain whether this relation is in 1NF. If not, decompose it to 1NF. [2 Points]

Already in 1NF. No need to decompose.

(ii) Explain whether the relation of no (i) is in 2NF. If not, decompose it to 2NF. [4 Points]

Not in 2NF due to having partial dependency.

2NF decomposition :

CAR_SALE_1 (Salesperson, Commission)

CAR_SALE_2 (Car, Salesperson, Date_sold, Discount_amt)

(iii) Explain whether the relation of no (ii) is in 3NF. If not, decompose it to 3NF. [4 Points]

Not in 3NF due to having transitive dependency.

3NF decomposition :

CAR_SALE_1 (Salesperson, Commission)

CAR_SALE_2_1 (Car, Salesperson, Date_sold)

CAR_SALE_2_2 (Date_sold, Discount_amt)

Just provided a short answer. You obviously need to draw the schema diagram with proper FDs..

Question 2 [CO5] : 10 Points

Consider the following relation:

T20CricketTournament (TournamentAcronym, Team_ID, Tournament_ID, Tournament_Title, Tournament_Matches, Base_Amount, Final_Amount, Team_Name, Team_Lead, Tournament_Year, Tournament_Sponsor, Tournament_Logo)

The primary key of the relation is underlined.

Suppose the following additional dependencies exist:

FD1: TournamentAcronym → Tournament_ID, Tournament_Title, Tournament_Matches, Base_Amount

FD2: Team_ID → Team_Name, Team_Lead

FD3: Tournament_ID → Tournament_Title, Tournament_Matches

Based on the given primary key,

(i) Explain whether this relation is in 1NF. If not, decompose it to 1NF. [2 Points]

Already in 1NF. No need to decompose.

(ii) Explain whether the relation of no (i) is in 2NF. If not, decompose it to 2NF. [4 Points]

Not in 2NF due to having partial dependency.

2NF decomposition :

T20CricketTournament_1 (TournamentAcronym, Tournament_ID, Tournament_Title, Tournament_Matches, Base_Amount)

T20CricketTournament_2 (Team_ID, Team_Name, Team_Lead)

T20CricketTournament_3 (TournamentAcronym, Team_ID, Final_Amount, Tournament_Year, Tournament_Sponsor, Tournament_Logo)

(iii) Explain whether the relation of no (ii) is in 3NF. If not, decompose it to 3NF. [4 Points]

Not in 3NF due to having transitive dependency.

3NF decomposition :

T20CricketTournament_1_1 (TournamentAcronym, Tournament_ID, Base_Amount)

T20CricketTournament_1_2 (Tournament_ID, Tournament_Title, Tournament_Matches)

T20CricketTournament_2 (Team_ID, Team_Name, Team_Lead)

T20CricketTournament_3 (TournamentAcronym, Team_ID, Final_Amount, Tournament_Year, Tournament_Sponsor, Tournament_Logo)